

Dr Jennifer Fitchett
Field Editor, International Journal of Biometeorology

Re: Submission ID 0cd18650-d5db-481f-b00b-bbde3cda74f7 — Major Revision

Dear Dr Fitchett,

Thank you for the opportunity to revise our manuscript. We are grateful to you and to both reviewers for the careful, constructive, and rigorous evaluation of our work, and we have addressed every comment individually and in detail.

In direct response to Reviewer 2's central request, we retained the originally submitted exposure (percentage of all households consisting of one person aged ≥ 65 years) and added prefectural ageing rate to the same model. The exposure coefficient changed from 656.2 (95% CI 326.9–985.6, $p < 0.001$) to 306.1 (95% CI –71.3–683.5, $p = 0.112$), a descriptive attenuation of 53.3%, and statistical significance was lost. Further adjustment for healthcare supply and official daily-maximum WBGT attenuated the estimate to near zero. We also independently and exhaustively reviewed the complete public file structure of the 10th NDB Open Data and confirmed that prefecture-by-age cross-tabulated counts for G004 are not publicly available.

These findings show that population age structure explains most of the ecological association we originally reported, an insight made possible only by the reviewers' request for this additional analysis. We have retitled the manuscript to foreground this finding, revised the Abstract, Introduction, Results, Discussion, and Conclusions accordingly, replaced the original simplified WBGT approximation with official measurements, removed the "socially blind" framing and the sixfold-comparison claim, renamed the GitHub repository, corrected the Miyatake et al. DOI, framed the study explicitly as hypothesis-generating, and substantially expanded the Limitations. A complete point-by-point response to every comment from the Editor, Reviewer 1 (35 items), and Reviewer 2 (6 items) is enclosed. We believe this demographically informed account of what drives the association is of direct value to the Journal's readers working on heat-health surveillance and social-vulnerability indicators.

We confirm that this submission has not been published elsewhere and is not under consideration by another journal. All authors have approved the revised manuscript.

Thank you for your consideration.

Sincerely,

Haruki Saito (corresponding author)
On behalf of all authors